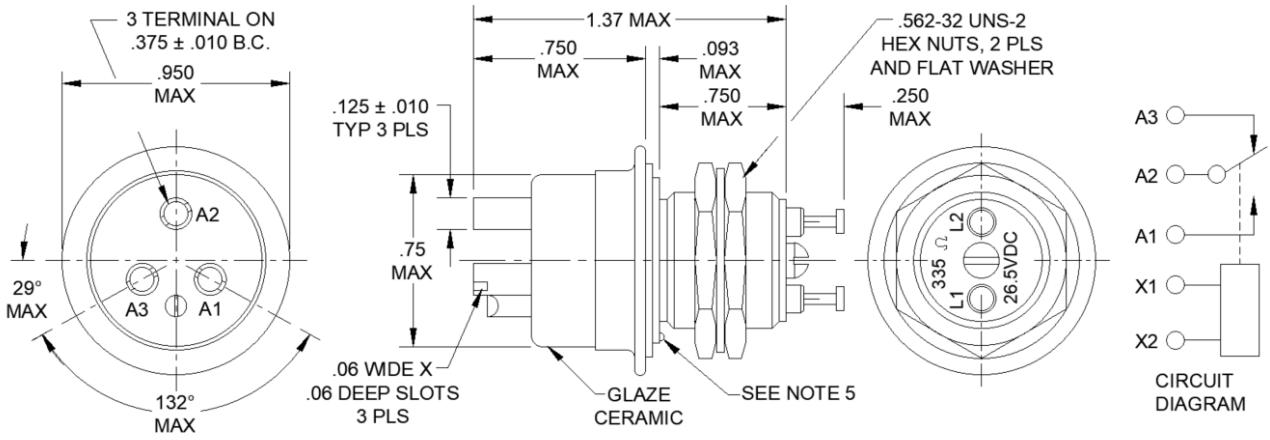
INCH-POUND
MIL-DTL-83725/5D
w/Amendment 1
24 September 2024
SUPERSEDING
MIL-R-83725/5D(USAF)
15 April 1994

DETAIL SPECIFICATION SHEET

RELAYS, VACUUM, SPDT, 16 AMPERES, RMS 3.5 KILOVOLTS PEAK (CARRY ONLY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83725.



Inches	mm
.020	0.51
.030	0.76
.060	1.52
.083	1.65
.125	2.36
.250	6.35
.520	13.21
.750	19.05
.950	24.13
1.37	34.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Part or Identifying Number (PIN) shall be permanently marked on the relay envelop.
5. A .030 X .060 X .020 maximum rectangular index slot shall be used to accommodate the index pin.

FIGURE 1. Relay outline and dimensions.

AMSC N/A FSC 5945

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

CONTACT DATA:

Configuration: SPDT.

Arrangement: One form C.

Load ratings:

Resistive: Carry only:

18 amperes dc or 60 Hz rms.
14 amperes rms, 2.5 MHz.
9 amperes rms, 16 MHz.
7 amperes rms, 32 MHz.

Rated operating voltage.

3.5 kV dc: 60 Hz or dc.
2.5 kV dc, 2.5 MHz.
2.0 kV dc, 16 MHz.
1.5 kV dc, 32 MHz.

Contact resistance:

Rated life:

Before: .01 ohm, maximum.
During: .100 ohm, maximum.
After: .100 ohm, maximum.

Contact bounce: Not applicable.

Capacitance:

2 picofarads, maximum across open contacts.
2.5 picofarads, maximum between open contacts and ground.

COIL DATA:

Duty rating: Continuous.

Maximum voltage: See table I.

Nominal voltage: See table I.

Pickup voltage: See table I.

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Dropout voltage: See table I.

Coil resistance: See table I.

Operate time: 8 milliseconds, maximum over the temperature range (or measured using 18.8 V dc coil voltage at 25° C).

Release time: 8 milliseconds, maximum over the temperature range (or measured using 18.8 V dc coil voltage at 25°C).

ELECTRICAL DATA:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and housing shall be 500 megohms or greater.

Dielectric withstanding voltage:

At atmospheric pressure:

Between all mated contacts in the open position: 5 kV dc peak.
Between high voltage terminals and housing: 5 kV dc peak.
Between coil and housing: 500 V rms, 60 Hz.

At reduced barometric pressure:

In accordance with MIL-STD-202-105, test condition C. Relays shall be mounted by normal mounting means. Points of application: Same as "at atmospheric pressure" except test voltage applied shall be 350 volts rms with other terminals grounded. Following the test, relays shall be examined for evidence of arcing, flashover, insulation breakdown, and damage.

ENVIRONMENTAL DATA:

Temperature range: -55°C to +125°C.

Vibration: MIL-STD-202- 204, test condition C (10 to 55 HZ, .06 inch (1.5 mm) double amplitude 55 to 2,000 Hz, 10 g's.

Shock: MIL-STD-202-213, test condition A, 50 g's, 11 milliseconds half sine.

PHYSICAL:

Terminal strength: 5 pounds pull.

Dimensions and configuration: See figure 1.

Termination: Solder terminal.

Weight: 1 ounce, maximum.

Marking: Applicable (circuit diagram need not be marked on relay).

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TABLE I. Operating characteristics. 1/

Dash number	Maximum voltage	Nominal voltage	Pickup voltage	Dropout voltage	Coil resistance	Pickup voltage over temperature range	Dropout voltage over temperature range
-001	32 V dc	26.5 V dc	14 V dc max	1.5 - 7 V dc	335 ohms ±10%	22 V dc	1 - 10 V dc

1/ At +25°C.

LIFE TEST REQUIREMENTS:

Mechanical cycling: 1 million cycles, 2 sample units (cycling rate, 36,000 per hour maximum; contact current shall exceed 10 milliamperes).

QUALITY ASSURANCE:

Dielectric withstanding voltage (for group A testing):

Tests to be conducted at atmospheric pressure rating only.

Duration of application: 5 to 10 seconds at a 10 percent increase in the dielectric withstanding voltage.

PIN: M83725/5-001.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-83725, this document references the following:
MIL-STD-202-105 MIL-STD-202-204 MIL-STD-202-213

Custodians:
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2024-046)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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